# CHANDU PRASAD BHAIRAPU



## AI Engineer

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## ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in LLM applications and agents, from multi-agent systems and RAG to AI evaluation and production deployment on AWS and Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in machine learning, time-series forecasting and Python development across enterprise and start-up environments.

## EXPERIENCE

### Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

**Kniit GmbH**

November 2025 – present Muldestausee, Germany

- Sole developer owning the full knowledge-extraction pipeline end-to-end – data ingestion, chunking, prompt engineering, and indexing (agentic, deterministic, and monolithic strategies) – reporting directly to the CTO and collaborating with SAP domain experts across the German energy sector.
- Designed, evaluated and optimized multi-agent LLM extraction algorithms (LangGraph: entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for knowledge extraction from complex regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% measured against an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- Delivered end to end as an AI-augmented engineer, from spec to production – leveraging Claude Code, custom skills, and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, LLM agents, LangGraph, LangChain, LiteLLM, MCP, RAG, GraphRAG, Vector Search, Azure, Docker, Kubernetes, FastAPI, React, Neo4j, MongoDB, Postgres**

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### GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

**Covestro Deutschland AG**

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application using LangGraph to optimize decision-making workflows for vector database queries, general inference, and summarization tasks.
- Developed a React-based front-end and a Python backend with FastAPI, leveraging Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, adhering to clean code practices. Version-controlled via GitLab, with CI/CD integration.
- **Tech stack: Python(Backend), LangGraph, LangChain, FastAPI, Docker, AWS, CDK, EC2, Lambda, GitLab CI/CD, React(Frontend), Postman**

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### GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

**Covestro Deutschland AG**

May 2024 – November 2024 Leverkusen, Germany

- Automated RAG's Generation and Retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.

- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- **Tech stack: Python, Scikit-Learn, PyTorch, TensorFlow, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS (S3, EC2, VPC, Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker, Streamlit, Vector Databases**

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### Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

**Covestro Deutschland AG**

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

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### Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

**Cognizant Technology Solutions**

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

## PROJECTS

**Project: Energy Customer Service Agent | TypeScript | Python | MCP | PostgreSQL | GitHub Actions**
- LLM agent that explains an energy bill through Python MCP tools (explain_bill, submit_meter_reading, hand-off_to_human), with per-customer access control, input validation and an eval set gated in CI.

## TECHNICAL SKILLSET

- **Programming:** Python, Git, GitLab CI/CD
- **AI / LLM:** LLM applications and agents (LangGraph, LangChain, LiteLLM), multi-agent systems, MCP, RAG, GraphRAG, prompt engineering, AI evaluation (RAGAS, PytrecEval), QLoRA fine-tuning, Hugging Face transformers
- **ML / Forecasting:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, time series (ARIMA, SARIMA, Prophet, LSTM), SQL, Postgres
- **Cloud and Deployment:** AWS (S3, EC2, Lambda, SQS, Load balancer, DynamoDB, VPC, CDK), Azure, Docker, Kubernetes, FastAPI, Postman, React(beginner), Agile
- **Databases:** MongoDB, Neo4j, Vector Databases (Qdrant, Pinecone, Azure AI Search)

## EDUCATION

**Master of Science in Artificial Intelligence** — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

**Minor in Computer Science & Engineering** — July 2019 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

**Bachelor of Technology in Mechanical Engineering** — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

## PUBLICATION & LANGUAGES

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles

German (B1) | English (Business Fluent) | Hindi (Fluent) | Telugu (Native)